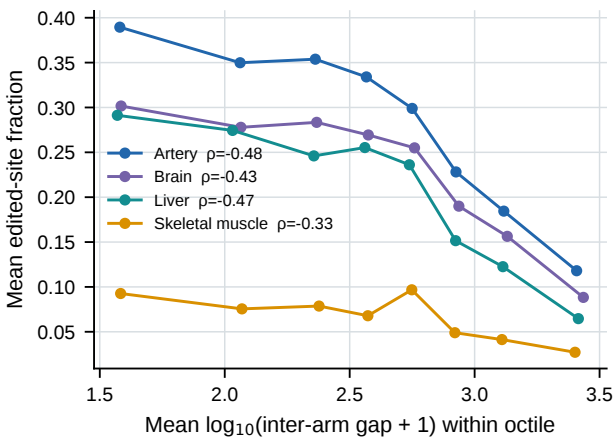
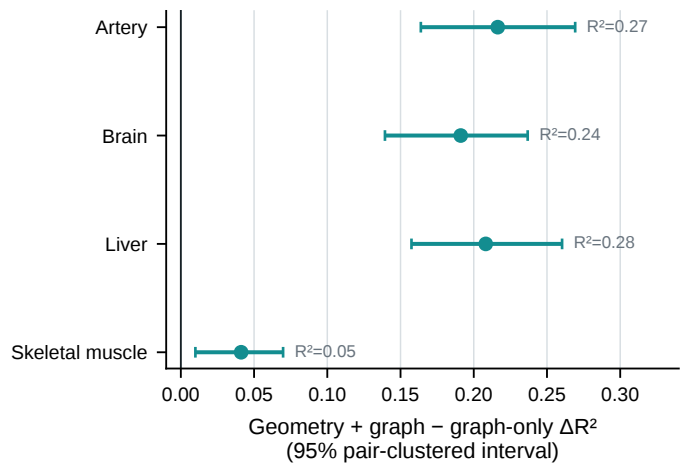
A

Evidence map: which information changes the recoverable signal?

INFORMATION LEVELS	Tested formulation	Outcome	Interpretation
	LEARNING & SUPERVISION		
LEARNING	Pair-first expanded supervision	● Selected	More labels; correct pair weight
	Continuous + ordinal auxiliary labels	▲ Limited	Small ranking gain
	Shared four-tissue training	▲ Limited	Transfer below preset criteria
	TARGET REPRESENTATION		
TARGET-LOCAL	Aligned target + helix-partner view	● Selected	Passes shifted-window control
	Focal node / contact geometry	▲ Limited	Localizes target effect
	Frozen RNA-language embeddings	✖ Not supported	F1-only; no ranking gain
	Substrate-wide all-A decoder	✖ Not supported	Stopped for futility
	Larger / deeper graph networks	✖ Not supported	No consistent benefit
	Graph Transformer / GATv2 expansion	+ Not pursued	Simpler validated model sufficient
	SUBSTRATE & CELLULAR CONTEXT		
SUBSTRATE / CONTEXT	Pair geometry / editing burden	◆ Supported†	Substrate propensity detectable
	Hierarchical engagement gate	▲ Limited	Ranking gain; unstable F1
	Bulk H3K36me3 overlap	▲ Limited	Liver AUPRC only
	RBP binding × bulk expression	✖ Not supported	No residual gain
	Transcript / repeat annotation panel	✖ Not supported	No site-level gain
	Matched donor/cell multimodality	■ Deferred	Matched data unavailable
	Matched in vivo structure / isoforms	■ Deferred	Matched data unavailable
	INTEGRATION		
INTEGRATION	Fixed whole-duplex + local-augmented logits	● Selected	Complementary predictions
	Unified end-to-end architecture	✖ Not supported	Below frozen ensemble

Comparators, effect sizes, and intervals: Table S2.

B



C